

Grade 9 lessons

Algebra: polynomials, factoring and quadratics

IN THIS BOOK

1. Polynomials
2. Factoring
3. Quadratic equations
4. Inequalities
5. Functions

HOW TO USE IT

Read one lesson together, then practice it. Each lesson matches 5 questions in the Grade 9 Kahoot and Blooket sets, and one stop on the Grade 9 trail in the Math Trail app.

Every lesson has the big idea, the steps, two worked examples, one mistake to watch for, and the words to know.

Polynomials

THE BIG IDEA

A polynomial is a sum of terms like $3x^2$, $5x$ and 7 . You simplify by combining like terms and expand by multiplying out brackets.

HOW TO DO IT

1. Like terms have the same letter and the same power. Add or subtract their numbers.
2. To expand $a(b + c)$, multiply a by every term inside.
3. To expand two brackets, multiply each term in the first by each term in the second: first, outer, inner, last.
4. Combine like terms at the end.

WORKED EXAMPLES

Simplify $3x + 5 + 2x - 1$.

x terms: $3x + 2x = 5x$. Numbers: $5 - 1 = 4$. Answer: $5x + 4$.

Expand $(x + 2)(x + 3)$.

$x \times x = x^2$. $x \times 3 = 3x$. $2 \times x = 2x$. $2 \times 3 = 6$. Combine: $x^2 + 5x + 6$.

WATCH OUT

x and x^2 are not like terms, so they cannot be added together. And $3(2x - 4)$ is $6x - 12$: multiply both terms.

WORDS TO KNOW

term

one piece of an expression

coefficient

the number in front of a letter

Factoring

THE BIG IDEA

Factoring is expanding in reverse. You rewrite an expression as things multiplied together.

HOW TO DO IT

1. First look for a factor every term shares, and take it out.
2. For $x^2 + bx + c$, find two numbers that multiply to c and add to b .
3. Write them in brackets: $(x + \text{first})(x + \text{second})$.
4. A difference of squares follows a pattern: $a^2 - b^2 = (a + b)(a - b)$.
5. Check by expanding your answer.

WORKED EXAMPLES

Factor $x^2 + 5x + 6$.

Two numbers that multiply to 6 and add to 5 are 2 and 3. Answer: $(x + 2)(x + 3)$.

Factor $x^2 - x - 6$.

Two numbers that multiply to -6 and add to -1 are -3 and 2. Answer: $(x - 3)(x + 2)$.

WATCH OUT

Watch the signs. If the last number is negative, the two numbers have different signs.

WORDS TO KNOW

factor

one of the things being multiplied

common factor

a factor shared by every term

Quadratic equations

THE BIG IDEA

If two things multiply to make zero, one of them must be zero. That fact lets you solve a quadratic equation by factoring.

HOW TO DO IT

1. Move everything to one side so the equation equals 0.
2. Factor the quadratic.
3. Set each bracket equal to 0 and solve.
4. For $x^2 = \text{a number}$, take the square root and remember both the positive and negative answers.
5. The discriminant $b^2 - 4ac$ tells how many real solutions: positive gives two, zero gives one, negative gives none.

WORKED EXAMPLES

Solve $x^2 - 5x + 6 = 0$.

Factor: $(x - 2)(x - 3) = 0$. So $x = 2$ or $x = 3$.

How many real solutions does $x^2 + 2x + 5 = 0$ have?

$b^2 - 4ac = 4 - 20 = -16$. It is negative, so there are no real solutions.

WATCH OUT

One side must be zero before you factor. And a quadratic usually has two answers. Give both.

WORDS TO KNOW

quadratic

an expression whose highest power is 2

discriminant

$b^2 - 4ac$

vertex

the turning point of the curve, at $x = -b \div (2a)$

Inequalities

THE BIG IDEA

An inequality is like an equation, but the answer is a whole range of numbers instead of just one.

HOW TO DO IT

1. Solve it step by step just like an equation.
2. If you multiply or divide both sides by a negative number, flip the inequality sign.
3. Read the answer: $x > 4$ means every number bigger than 4.
4. Check by trying one number from your answer.

WORKED EXAMPLES

Solve $2x + 3 > 11$.

Subtract 3: $2x > 8$. Divide by 2: $x > 4$.
Check with 5: $2 \times 5 + 3 = 13$, which is more than 11.

Solve $-3x < 12$.

Divide by -3 and flip the sign: $x > -4$.

WATCH OUT

Flip the sign only when you multiply or divide by a negative. Adding or subtracting a negative does not flip it.

WORDS TO KNOW

inequality a statement using $<$, $>$, $\leq$ or $\geq$

Functions

THE BIG IDEA

A function is a rule. You put a number in and get exactly one number out. $f(x)$ is read "f of x".

HOW TO DO IT

1. To find $f(3)$, replace every x in the rule with 3 and calculate.
2. Follow the order of operations: powers before multiplying.
3. The x -intercept is where $y = 0$. Set the rule equal to 0 and solve.
4. In a table, if y changes by the same amount each time x goes up 1, the rule is a line and that amount is the slope.

WORKED EXAMPLES

$f(x) = 2x^2 - 1$. Find $f(3)$.

$$3^2 = 9. 2 \times 9 = 18. 18 - 1 = 17.$$

Where does $y = 2x - 6$ cross the x -axis?

Set $y = 0$: $2x - 6 = 0$, so $2x = 6$ and $x = 3$.

WATCH OUT

$f(3)$ does not mean f times 3. And in $2x^2$, square first, then multiply by 2.

WORDS TO KNOW

function

a rule that gives one output for each input

x -intercept

where a graph crosses the x -axis